Practice Quiz: The Invention Roadmap

Part A: Multiple Choice

1. Expected free energy in Active Inference planning consists of: A) Revenue and cost estimates B) Pragmatic value (progress toward goals) and epistemic value (reduction of uncertainty) C) Speed and quality metrics D) Team size and budget projections

2. Staged development in invention planning means: A) Building the entire product in one stage and then testing B) Breaking the project into phases where each phase generates information for decisions in the next phase, deferring irreversible decisions until uncertainty is reduced C) Having many stages but making all decisions at the beginning D) Staging a dramatic product launch

3. An inventor who invests heavily in product packaging before confirming that the core mechanism works is committing which planning error? A) Premature optimization — investing in low-uncertainty aspects while high-uncertainty questions remain unresolved B) Excessive caution — they should invest even more in packaging C) Staged development — they are properly sequencing their work D) Adaptive planning — they are adjusting to market feedback

4. The planning horizon is: A) The date when the project will be completed B) The temporal distance at which the generative model can still make useful predictions — beyond which detailed planning is counterproductive C) The distance to the nearest competitor D) The maximum height of the product

5. Option value in invention planning refers to: A) The value of stock options in the company B) The benefit of keeping multiple approaches open before committing, because the information gained may outweigh the cost of maintaining parallel paths C) The number of options on a product menu D) The value of optional features in the product

6. SpaceX developed rockets through a staged process (Falcon 1, Falcon 9, reusable operations) rather than building a fully reusable rocket immediately. This strategy: A)

Wasted time on simpler rockets when they could have built the final version directly B) Minimized expected free energy by resolving fundamental uncertainties before committing to complex, expensive designs C) Was forced by government regulation D) Was accidental — they did not have a plan

7. Pivot criteria are: A) The angle at which the product rotates B) Pre-specified conditions under which the plan would fundamentally change, written before emotional investment clouds judgment C) Criteria for hiring new team members D) Standards for product quality

Part B: Short Answer and Analysis

1. An inventor has developed a working prototype of a new type of portable water filter. The prototype works in laboratory conditions but has not been tested in real-world conditions, with real users, or at manufacturing scale. Design a three-stage plan for the next phase of development. For each stage, specify: (a) the key uncertainty to resolve, (b) the actions to take, (c) the go/no-go criteria, and (d) how the learning from this stage would update the plan for the next stage.

2. An inventor is choosing between two project paths:

3. **Path A**: A low-risk incremental improvement to an existing product (high pragmatic value, low epistemic value — they already know it will work, but the upside is modest).

4. **Path B**: A high-risk novel approach that could be transformative if it works but has major technical uncertainty (lower pragmatic value currently, but high epistemic value — testing will resolve critical unknowns). Using the expected free energy framework, analyze which path the inventor should pursue. Under what conditions would Path A be preferable? Under what conditions would Path B be preferable? Is there a way to pursue both that leverages option value?

5. Review the microwave oven case study: Percy Spencer's discovery of microwave heating in 1945 led to a consumer product only in the late 1960s. Identify three points in this 20-year roadmap where the plan had to adapt based on prediction errors. For each point, describe: (a) the prediction error (what was expected vs. what happened), (b) the

plan revision (how the roadmap changed), and (c) whether this was a parameter adjustment or a structural change to the plan.